

Cost-aware graph augmentation for robust networks via effective resistance

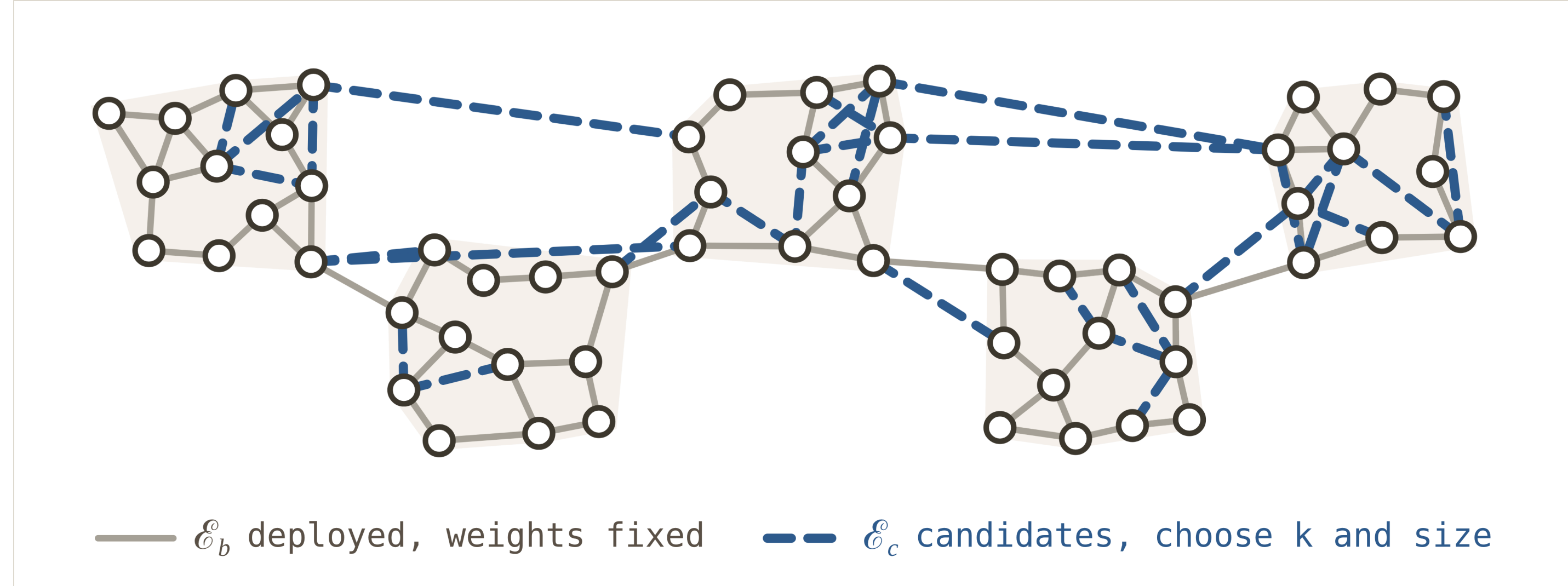
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01 Optimizing a pre-deployed network

Total effective resistance R_{tot} (the Kirchhoff index) is a tractable spectral proxy for robustness. Reducing it on a *deployed* weighted network requires deciding **which links to add and how strongly to weight them**. This couples a discrete selection problem with a continuous sizing problem under a single global budget. Costs vary across candidate links, combining a fixed installation overhead with a length-dependent penalty.



The setup. We select exactly k candidates and size each within $[w_{\min}, w_{\max}]$. Every candidate carries its own unit price c_i and the total spend cannot exceed C .

OBJECTIVE

$$\mathbf{L}(\mathbf{w}) = \mathbf{L}_b + \underbrace{\sum_{e \in \mathcal{E}_c} w_e \mathbf{a}_e \mathbf{a}_e^\top}_{\mathbf{L}_a(\mathbf{w})}, \quad \mathbf{M}(\mathbf{w}) := \mathbf{L}(\mathbf{w}) + \frac{\mathbf{1}\mathbf{1}^\top}{n} \succ 0$$

Here $\mathbf{L}_b$ is the base Laplacian and $\mathbf{L}_a(\mathbf{w})$ the augmentation Laplacian, where candidate e enters through its incidence vector $\mathbf{a}_e$.

$$R_{\text{tot}}(\mathbf{w}) = n \operatorname{Tr}[\mathbf{M}^{-1}(\mathbf{w})] - n$$

02 Joint selection and sizing

MIXED-INTEGER PROGRAM

minimize $R_{\text{tot}}(\mathbf{w})$ robustness of the upgraded network

subject to $\mathbf{c}^\top \mathbf{w} \leq C$ one global budget

$z_i w_{\min} \leq w_i \leq z_i w_{\max}$ perspective box, $z_i = 0 \Rightarrow w_i = 0$

$\mathbf{1}^\top \mathbf{z} = k, \quad z_i \in \{0, 1\}$ exactly k new links

(P₀)

03 SDP relaxation and feasible recovery

The binary variables z_i render (P₀) NP-hard, and evaluating the inverse in R_{tot} requires an $\mathcal{O}(n^3)$ matrix factorization. Relaxing $\mathbf{z}$ to $[0, 1]^p$ removes the combinatorial difficulty. Since $\mathbf{M}(\mathbf{w}) \succ 0$ for a connected base graph, $\operatorname{Tr}(\mathbf{M}^{-1}(\mathbf{w})) \leq \operatorname{Tr}(\mathbf{R})$ can be imposed through a Schur complement, eliminating the explicit inverse and yielding a convex formulation.

SEMIDEFINITE SURROGATE

minimize $n \operatorname{Tr}(\mathbf{R}) - n$

subject to $\begin{bmatrix} \mathbf{M}(\mathbf{w}) & \mathbf{I} \\ \mathbf{I} & \mathbf{R} \end{bmatrix} \succeq 0$

$\mathbf{c}^\top \mathbf{w} \leq C$

$z_i w_{\min} \leq w_i \leq z_i w_{\max}, \quad i = 1, \dots, p$

$\mathbf{1}^\top \mathbf{z} = k, \quad 0 \leq z_i \leq 1, \quad \mathbf{R} \in \mathbb{S}_+^n$

(P₁)

Rounding and repair. Keep the k largest entries of $\mathbf{z}^*$ and break ties toward cheaper links. Project the weights onto the induced box. If the design is over budget, scale toward $w_{\min} \hat{\mathbf{z}}$ until $\mathbf{c}^\top \hat{\mathbf{w}} \leq C$. The recovered design is feasible only when the selected support can carry the minimum weights.

Proposition 1. Assume the base graph is connected, so $\mathbf{M}(\mathbf{w}) \succ 0$ for all feasible $\mathbf{w}$. Then (P₁) satisfies $R_{\text{tot}}^{\text{SDP}} \leq R_{\text{tot}}^{\text{MI}}$. Moreover, at any optimum $(\mathbf{R}^*, \mathbf{w}^*, \mathbf{z}^*)$ of (P₁), $\mathbf{R}^* = \mathbf{M}^{-1}(\mathbf{w}^*)$. Consequently, the exact optimum of (P₁) is a lower bound on (P₀).

Remark 1. Given a feasible rounded pair $(\hat{\mathbf{w}}, \hat{\mathbf{z}})$, we have

$$R_{\text{tot}}^{\text{SDP}} = \operatorname{val}(\mathcal{P}_1) \leq \operatorname{val}(\mathcal{P}_0) = R_{\text{tot}}^{\text{MI}} \leq R_{\text{tot}}(\hat{\mathbf{w}}) =: R_{\text{tot}}^{\text{RND}}$$

so the relative ratio $(R_{\text{tot}}^{\text{RND}} - R_{\text{tot}}^{\text{SDP}}) / R_{\text{tot}}^{\text{SDP}}$ is an a posteriori gap certificate for the recovered design.

04 Solved as a cone program

Scaled vectorization stacks the unknowns as $\mathbf{x} = [\operatorname{svec}(\mathbf{R})^\top \mathbf{w}^\top \mathbf{z}^\top]^\top$. The constraints take the standard conic form $\mathbf{F}\mathbf{x} + \mathbf{s} = \mathbf{b}$, with $\mathbf{s} \in \mathcal{K}$.

$$\mathcal{K} = \underbrace{\mathbb{S}_+^{2n}}_{\text{LMI}} \times \underbrace{\mathbb{R}_+^p}_{\text{budget}} \times \underbrace{\mathbb{R}_+^p \times \mathbb{R}_+^p \times \mathbb{R}_+^p \times \mathbb{R}_+^p}_{\text{perspective box and } z \text{ bounds}} \times \underbrace{\{0\}}_{\text{cardinality}}$$

- Solved by ADMM on the homogeneous self-dual embedding, so the reported bound holds to the solver tolerance.
- A **cached LDL^T factorization** reduces each affine step to two triangular solves.

05 Up-to- k greedy with rank-one updates

A cost-aware greedy algorithm adds up to k links, stopping early if no remaining candidate is affordable. The Sherman–Morrison rank-one update provides each marginal gain in closed form. The inverse $\mathbf{M}^{-1}$ is formed once and then updated after every insertion rather than recomputed. Each trial weight is priced by a **tentative budget allocation** $\mathbf{s} = C_{\text{rem}}/k_{\text{rem}}$.

MARGINAL GAIN, EXACT

$$\Delta R_{\text{tot}} = \frac{n w_{(v_i, v_j)} b^2(v_i, v_j)}{1 + w_{(v_i, v_j)} r(v_i, v_j)}$$

$r(v_i, v_j) := \mathbf{a}^\top \mathbf{M}^{-1} \mathbf{a}$ $b^2(v_i, v_j) := \|\mathbf{M}^{-1} \mathbf{a}\|_2^2$ total cost: $\mathcal{O}(n^3 + k(pn + n^2))$

TRIAL WEIGHT

$$w_e := \max\{w_{\min}, \min\{w_{\max}, s/c_e\}\}$$

from the tentative allocation $\mathbf{s} = C_{\text{rem}}/k_{\text{rem}}$

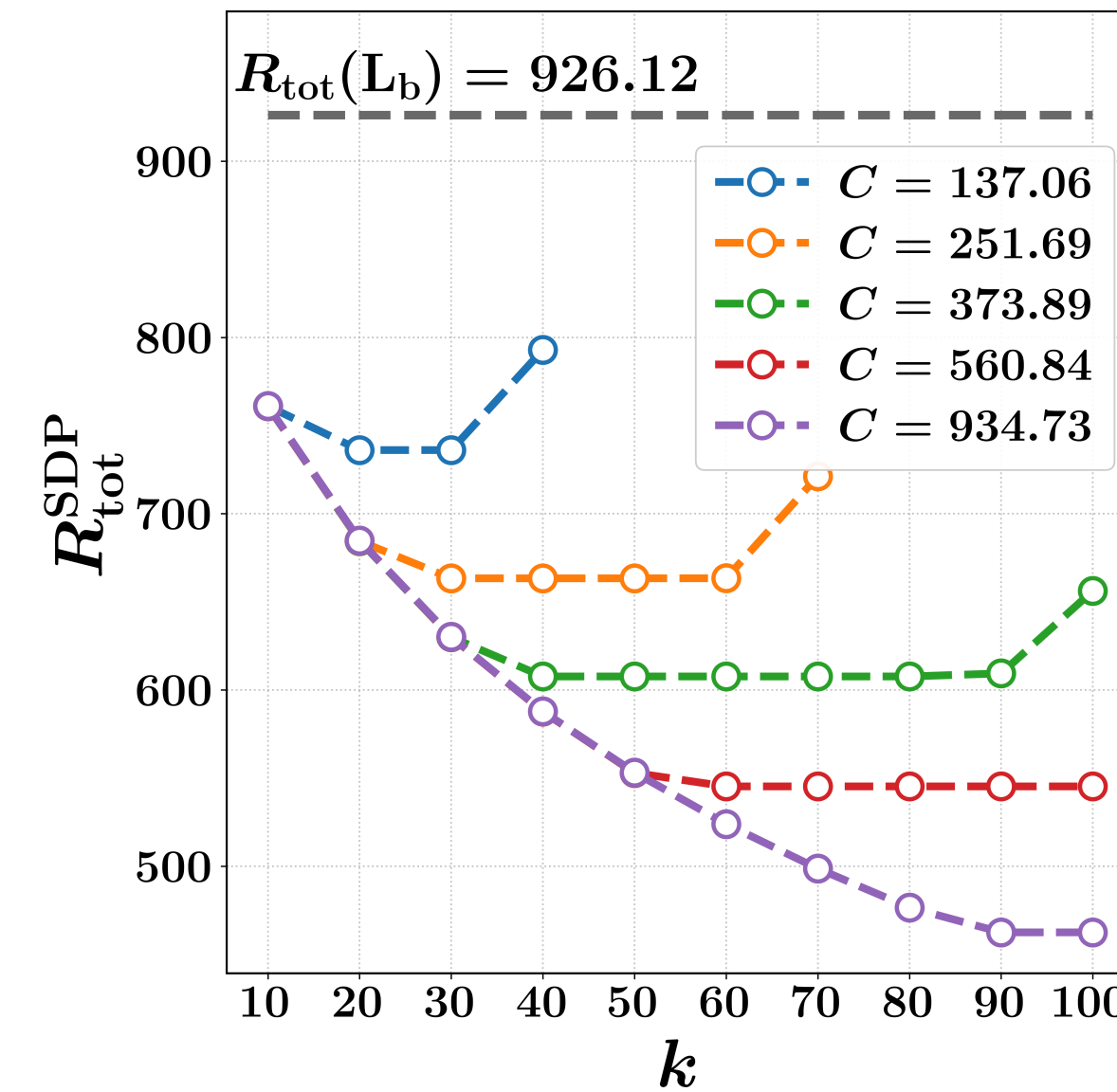
06 Instances and candidate sets

Synthetic. Base graph of $n = 100$ planar vertices with lognormal weights. Candidate weights span the 0.75 to 0.95 quantiles. Three k -means regions give $k_{\text{in}} = 3$ neighbour links and a sparse backbone.

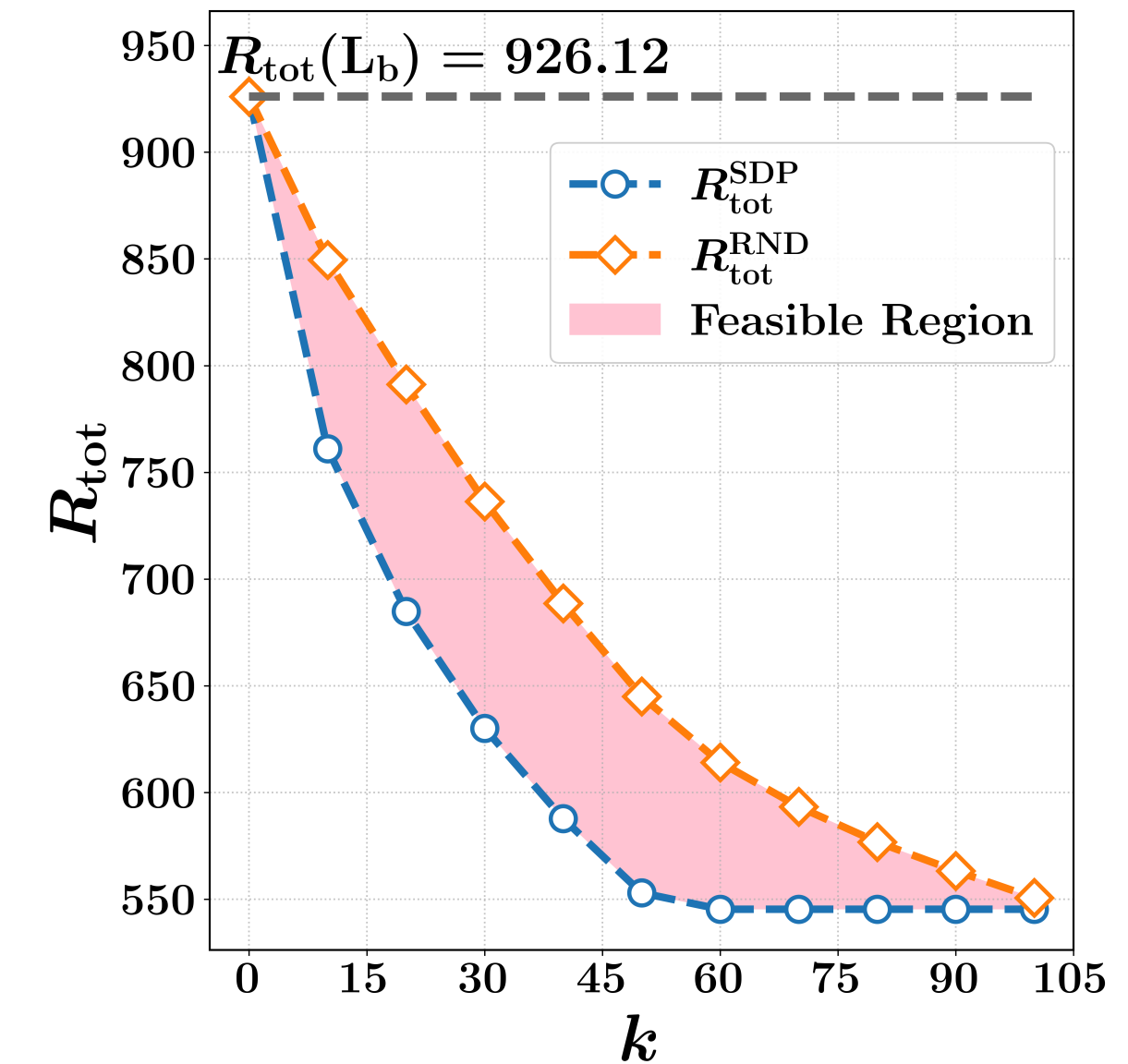
Cost model. Candidate edges follow $\mathbf{c} = c_{\text{fixed}} \mathbf{1} + \beta \mathbf{d}_c$, so each candidate has a per-unit price. Here, c_{fixed} is the baseline per-unit cost, and the candidate link lengths $\mathbf{d}_c$ are normalized.

Budget regimes. Budgets are deduced from the candidate cost structure. The least budget admitting k links is $C_{\min}(k) = w_{\min} \sum_{i=1}^k c_{(i)}$, summed over the cheapest candidates. Five levels centre on the anchor $C_{\min}(k_{\max}) = 373.89$, two at the thresholds $k_{\max}/3$ and $2k_{\max}/3$ and two at $1.5\times$ and $2.5\times$.

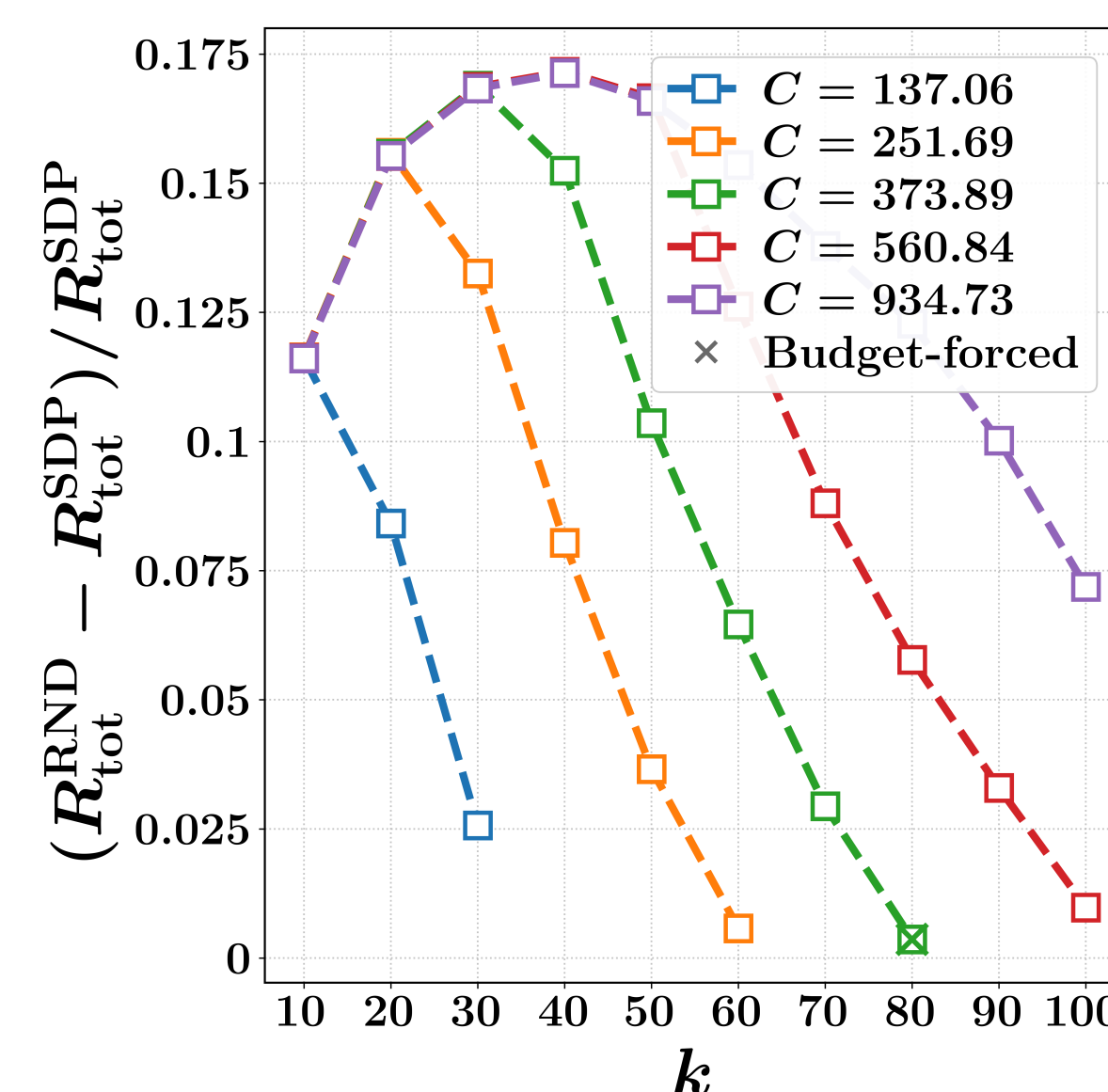
07 What the budget buys



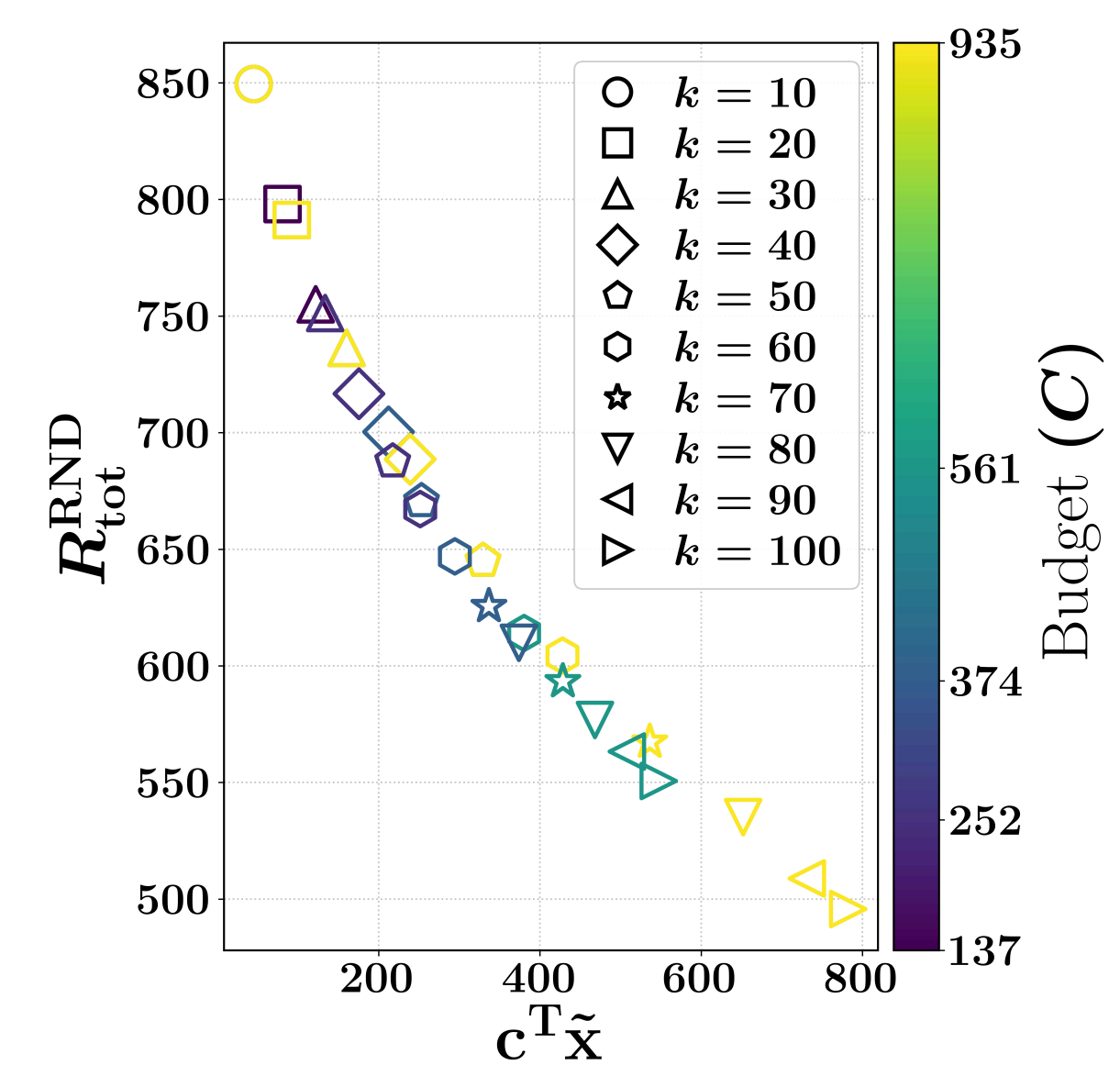
1a. $R_{\text{tot}}^{\text{SDP}}$ versus k at different budgets.



1b. Feasible region between the bound and the rounded design.

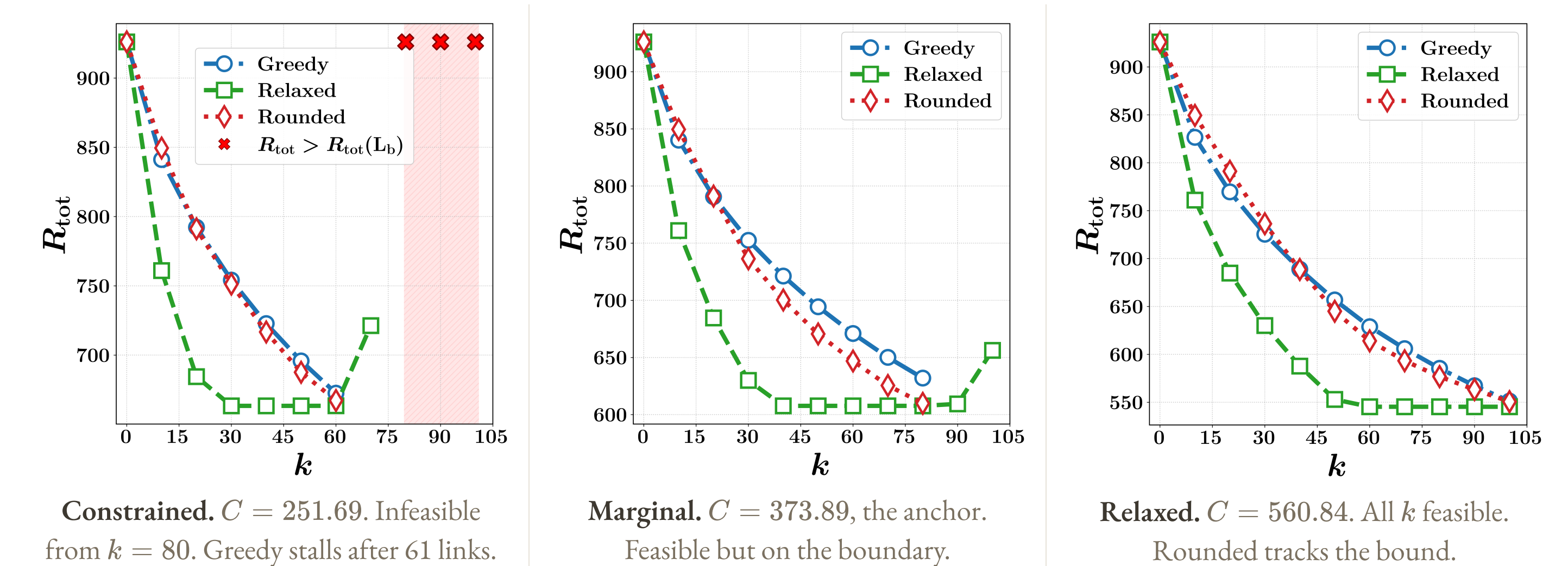


1c. Relative gap $(R_{\text{tot}}^{\text{RND}} - R_{\text{tot}}^{\text{SDP}}) / R_{\text{tot}}^{\text{SDP}}$ at each budget.

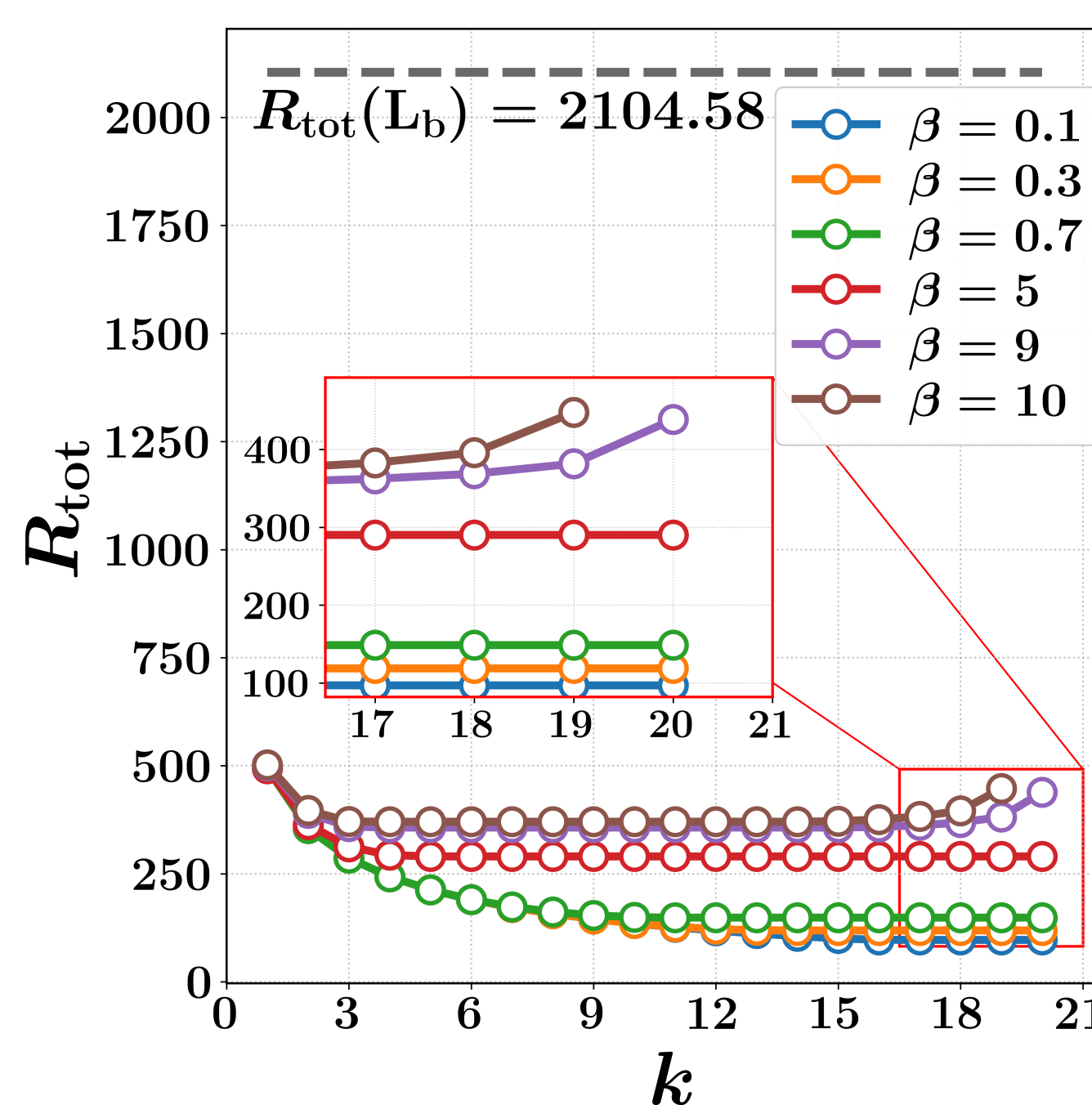


1d. Pareto front against budget spent at fixed k .

08 Budget regimes



09 California road network



2a. $n = 500$ sampled nodes at $C = 150$, sweeping β . A steeper β lifts the curve.

10 Takeaways

- The exact SDP optimum lower-bounds every feasible discrete augmentation.
- The Pareto front exposes the trade-off between effective resistance and budget spend.
- The relaxed-to-rounded ratio is an a posteriori gap certificate for each design.
- The up-to- k greedy trades solution quality for computational speed.

Key references. Ghosh, Boyd & Saberi, *SLAM Rev.* 2008 · Kooij & Achterberg, *Oper. Res. Lett.* 2023 · Predari et al., *SNAM* 2023 · O'Donoghue et al., *JOTA* 2016 · Leskovec et al., *Internet Math.* 2009

Notation. Scalars are lowercase and vectors bold lowercase. Matrices are bold uppercase. $\mathbf{A} \succeq \mathbf{B}$ means $\mathbf{A} - \mathbf{B}$ is positive semidefinite. $\mathbb{S}_+^n$ is the symmetric PSD cone and $\operatorname{Tr}(\cdot)$ the trace. A superscript $*$ marks optimal variables.

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Paper & code
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